DentOS

Clinical Suite — Build Progress & Strategic Assessment

Dental-clinical and project-delivery review · 21 June 2026

DentOS is a staff-only dental clinic management system for Flutter Desktop: an encrypted, offline-first local database as the source of truth, with cloud sync, multi-branch support, and an RSA-signed licensing system. This document is a single read-out of **what has been built**, **what remains**, and an honest professional assessment — written through both a practising-dentist lens and a project-delivery lens — of where it stands against industry standards and its real target market: small-to-mid private clinics in Pakistan and the wider region.

Current readiness in one line: an excellent scheduling, charting and basic-billing product that a single-doctor clinic could pilot today — not yet a full clinical system-of-record. The distance to that is well understood and listed below.

01 What's been built

PLATFORM & INFRASTRUCTURE

- **Encrypted local database.** Drift database (SQLCipher-ready) as the offline source of truth, at schema version 8 with a working migration path.
- **App shell and routing.** go_router StatefulShellRoute across 8 modules; dark sidebar + light canvas in the “Futuristic Clinical Minimalist” theme via a DentColors theme extension.
- **Licensing and lock.** RSA-signed offline license, a 48-hour offline gate (all tiers), and a monotonic anti-rollback clock.
- **Design system.** Sora / Manrope / JetBrains Mono, KPI cards, panels, status chips, segmented controls and reusable fields used consistently across screens.

CLINICAL

- **FDI odontogram.** A 2D dental chart with tap-to-set tooth conditions, backed by per-tooth records.
- **Treatment plans.** Staged treatment plans with steps and status, surfaced on the patient record.
- **Patient detail screen.** A full drill-down: quick stats, last visit, personal details, visit history, invoices, treatment plan and a dental-chart card.
- **3D tooth viewer.** A rotate-only 3D model viewer (three_js) toggled alongside the 2D chart, which remains the editable clinical tool.
- **Treatment catalogue.** A configurable procedure catalogue with pricing and duration.

FRONT OFFICE — APPOINTMENTS & PATIENTS

- **Appointments.** Day agenda, mini-calendar, Quick Book drawer, and a New Appointment dialog with a live patient combobox and atomic create-or-book in a single transaction.
- **Quick actions.** Per-appointment Arrived (with a confirm dialog) and Bill (opens a pre-filled invoice), plus a persistent Billed state per appointment.
- **Patients.** Searchable patient list with add/edit and a drill-down detail view.

- **Context-aware search and alerts.** A global search that targets the current screen's data, and a live notification bell with real counts for low stock, unpaid invoices and recall-due patients.

BILLING, INVENTORY & REPORTS

- **Billing.** Invoices with line items, insurance adjustment and status; invoice PDF / print; billing KPIs.
- **Inventory.** Stock tracking with par / reorder levels and low-stock alerts.
- **Reports.** Analytics (fl_chart): revenue trend, procedure mix and per-dentist performance.
- **Live dashboard.** KPIs, today's schedule, weekly revenue, top procedures and clinic status on efficient aggregate streams, with Today / Week / Month scoping.

CLOUD, SYNC, LICENSING & ONBOARDING

- **Self-service onboarding.** An Edge Function verifies the signed license, provisions a Supabase auth user tagged with the clinic, and upserts the clinic row — driven from the setup wizard.
- **Full cloud sync.** A last-write-wins sync engine by UUID across all tables, with an owned-children strategy for tooth records, plan steps and invoice items, and foreign-key remapping; manual sync trigger; heartbeat right after setup.
- **Multi-branch and staff.** Premium branches, seat-based staff and branch-scoped logins.
- **Auth screens.** Glassmorphic login / setup / locked / reconnect screens with a looping video background.

02 What remains — priority snapshot

Grouped by urgency. **Blocker** = fix before a clinic relies on it; **Core** = needed to be a true clinical record; **High** = competitive and adoption; **Medium** = quality and operations.

AREA	ITEM	PRIORITY
Platform	Windows support (target market runs Windows)	Blocker
Data	Migration safety harness + schema-history reconcile	Blocker
Billing	Partial payments, patient ledger, day-sheet, receipts	Blocker
Data	Proven backup / restore + disaster recovery	Blocker
Sync	Conflict handling (last-write-wins drops edits)	Blocker
Clinical	Progress / SOAP clinical notes	Core
Clinical	Periodontal charting (pocket depths, BOP, recession)	Core
Clinical	Imaging — store / view X-rays + intraoral photos	Core
Clinical	Medical history, medications, ASA + contraindication flags	Core
Clinical	Prescriptions and signed consent forms	Core
Clinical	Dental lab case tracking	Core
Growth	Appointment reminders (WhatsApp hand-off)	High

AREA	ITEM	PRIORITY
Growth	Recall / recare automation	High
Growth	Data import / onboarding from old records	High
Growth	Owner reports: A/R ageing, production vs collection	High
Eng	Automated tests (sync, migrations, billing maths)	High
Release	RSA modulus, deploy function, SQLCipher native wiring	High
Eng	Crash / error reporting + structured logging	Medium
Eng	Role-based access enforcement audit (data layer)	Medium
Eng	In-app auto-update; bundle fonts; macOS entitlement	Medium

03 Where DentOS is genuinely strong

- **Local-first, encrypted core.** An encrypted database that runs fully offline is a real advantage where power and connectivity are unreliable — the clinic keeps working when the internet does not.
- **Cloud sync and multi-branch.** Sync with row-level security and per-branch scoping gives a credible path from one practice to a multi-branch group.
- **Monetisable licensing.** The RSA-signed license and automated provisioning make this something you can actually sell and control, not just an internal tool.
- **Modern interface.** A clean, current UI is a genuine differentiator against both the dated global incumbents and the crude local alternatives.
- **Charting foundation.** An FDI odontogram plus a 3D model viewer is a strong, marketable starting point.

04 Deployment blockers

These cause data loss, money errors, or “we can’t even install it” — they come first.

- **Windows support.** Most clinics in the target market run Windows; a macOS-only build limits the addressable market to a fraction of it. Highest-impact item for real-world reach.
- **Migration safety.** The schema-version history needs reconciling, and a harness that upgrades old databases through every version is essential before any clinic depends on it — a bad migration on live patient data is catastrophic.
- **Payments depth.** Real billing needs partial payments, a per-patient ledger with running balance, a daily cash close, printable receipts, and local methods (cash, EasyPaisa, JazzCash). Money must reconcile to the rupee and be auditable.
- **Proven backup and recovery.** Backups are only real once a restore has been shown to rebuild a working clinic; add tested restore and an offsite path.
- **Sync conflict handling.** Last-write-wins silently drops the losing edit; with two front-desk users on the same record that is quiet data loss — detect and surface conflicts, especially on financial rows.

05 Clinical gaps

Today it is a strong front-office and restorative-charting system. To be a clinical system-of-record it needs:

- **Progress / SOAP notes.** Structured, templated per-visit notes are a medico-legal necessity and the backbone of a clinical record.
- **Periodontal charting.** Six-point pocket depths, bleeding on probing, recession and mobility tracked over time — standard of care, currently absent.
- **Medical history.** Medications, systemic conditions and ASA status with contraindication warnings, extending the existing allergy field.
- **Imaging.** Store and view radiographs and intraoral photos against the patient and tooth — today X-rays are billed but not kept.
- **Prescriptions and consent.** Printable prescriptions and digitally stored, signed consent forms.
- **Lab case tracking.** Crowns, dentures and implants go out to a lab with due dates — a daily workflow the catalogue already implies.

06 Operational, growth & engineering

ADOPTION & GROWTH

- **Appointment reminders.** No-shows are the largest avoidable revenue leak; WhatsApp reminders are the market norm, and even a manual pre-filled hand-off would cut them materially.
- **Recall automation.** The recall-due status exists; close the loop with a worklist and reminders to recover revenue.
- **Data import and owner reports.** An importer for existing patients removes the biggest switching barrier; A/R ageing, production-vs-collection and provider productivity are the reports an owner buys the software to see.
- **Updates and localisation.** A safe in-app update path, Urdu labels for front-desk staff, and any required tax fields on invoices.

ENGINEERING & PROCESS

- **Automated tests.** Coverage for sync, migrations and billing maths is not optional for medical and financial records.
- **Reliability tooling.** Centralised crash / error reporting and structured logs so field failures are visible.
- **Access-control audit.** Confirm roles gate clinical and financial data at the data layer, not merely in the UI.
- **Scale and retention.** Validate performance at ten-thousand-plus patients and several years of data; make long-term record retention an explicit rule (the soft-delete and audit log already support it).

07 Suggested phased roadmap

PHASE	FOCUS	KEY ITEMS
Phase A	**Harden for deployment**	Windows build · migration harness + schema reconcile · payments depth (partial, ledger, day-sheet) · proven backup/restore · sync-conflict surfacing
Phase B	**Complete the clinical record**	SOAP notes + medical history · periodontal charting · imaging storage · prescriptions + consent · lab case tracking
Phase C	**Drive adoption & growth**	Appointment reminders · recall automation · data import · owner reports (A/R ageing) · Urdu + tax fields
Phase D	**Quality & scale**	Automated tests · crash/error reporting · RBAC audit · load testing · in-app auto-update

08 Real-world verdict

The opportunity is real and specific: affordable, modern, offline-resilient, encrypted, multi-branch software for clinics the global vendors price out and the local vendors serve poorly. As it stands, DentOS is an excellent scheduling, charting and basic-billing product — a single-doctor clinic could pilot it today. It is not yet a full system-of-record, and the distance to that is gated by five things: **Windows support, payments and ledger depth, clinical and periodontal notes, appointment reminders, and migration safety.** Close those, and a multi-chair or multi-branch practice can confidently run on it.